CE 310 — Week 10 Weekly Quiz

Week: 10 | **Topic:** Multiple Linear Regression — Concepts and Python/statsmodels

Due: Closes **8:00 AM Tuesday, Nov 3** — the day before Exercise 10 and Homework 10 are due. Taken on D2L
Points: 15 questions × 1 pt each = 15 pts
Attempts: Two — the higher score counts
Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Python tutorial — multiple linear regression, adjusted R^2, dummy variables, multicollinearity, interaction terms, and the Python tools that go with them. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. Two models are fitted to the same dataset. Model 1 uses three predictors and reports $R^2 = 0.643$ with **AIC = 71,240**. Model 2 adds a fourth predictor, raising R^2 to 0.646 while **AIC rises to 71,262**. Which conclusion does AIC support?

- A) Model 2 is better, because its R^2 is higher
- B) The two models cannot be compared, because they use different numbers of predictors
- C) AIC and R^2 disagree here, so neither model can be selected
- D) Model 1 is better — the fourth predictor did not earn the complexity it cost

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Q2. In Week 9, outdoor temperature explained $R^2 = 0.601$ of the variance in cooling energy — leaving 39.9% unexplained. Which of the following physical factors is the **most likely** additional predictor that could explain some of the remaining variance?

- A) The building's street address
- B) Whether the building is occupied during business hours
- C) The year the building was constructed
- D) The color of the building's exterior walls

Q3. **Adjusted R^2** (Adj. R^2) differs from R^2 in that it:

- A) Measures the fraction of variance explained by the model, scaled to the range 0–100%
- B) Penalizes added predictors that do not improve fit, so it can decrease
- C) Equals the square of the Pearson r between observed and predicted values
- D) Is always larger than R^2 because it corrects for sample size bias

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Q4. [True/False] A 0/1 dummy variable added to a regression shifts the intercept for one category without changing the slope on the continuous predictor.

Q5. The Model 2 regression equation is: `Cooling_kWh = −56.87 + 1.026 × Outdoor_Temp_F + 17.85 × BH`. **Without running any code**, compute the predicted cooling load for an occupied hour (BH = 1) at Outdoor_Temp_F = 90°F.

- A) 17.85 kWh/hr
- B) 35.47 kWh/hr
- C) 110.19 kWh/hr
- D) 53.32 kWh/hr

Q6. After fitting the two-predictor model `Cooling_kWh ~ Outdoor_Temp_F + BH`, the regression output shows: R^2 = **0.697**, BH coefficient = **17.9 kWh/hr**, p-value for BH ≈ 0. Which interpretation is correct?

- A) Adding occupancy (BH) raises explained variance from 60.1% to 69.7%, and occupied hours need about 18 kWh/hr more cooling
- B) R^2 increased from 0.601 to 0.697 primarily because adding any variable always increases R^2; the BH coefficient is not meaningful
- C) The BH coefficient of 17.9 kWh/hr represents the mean cooling demand during business hours, not an incremental effect
- D) R^2 = 0.697 means the model predicts cooling energy with 69.7% accuracy; the remaining 30.3% of predictions are incorrect

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Q7. [Select all that apply] Two predictors in a multiple regression are strongly correlated with each other. Which of the following follow?

- (A) The coefficient standard errors inflate, so the individual coefficients become unstable
- (B) The model's overall predictions become unusable
- (C) It becomes hard to attribute an effect to one predictor rather than the other
- (D) R^2 necessarily drops
- (E) Adding the second predictor buys little additional explained variance

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Q8. An **interaction term** between two predictors x_1 and x_2 is created by multiplying them together ($x_1 \times x_2$). What does a significant interaction term tell you?

- A) That the effect of x_1 on the outcome changes depending on the value of x_2
- B) That the effect of x_1 on the outcome is the same regardless of the value of x_2
- C) That x_1 and x_2 are collinear and should not both be included in the model
- D) That x_1 and x_2 together explain 100% of the variance in the outcome

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Q9. [Matching] Match each quantity in a regression output to what it reports.

Choices:

- (A) R^2
- (B) Adjusted R^2
- (C) AIC
- (D) The p-value on a coefficient
- (E) A coefficient

Matches:

1. The change in the outcome per unit change in that predictor, holding the others fixed
2. The fraction of outcome variance the model explains
3. The strength of the evidence that this predictor's effect is not zero
4. The explained fraction, penalised for the number of predictors used

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Q10. [Ordering] You want to know whether occupancy is worth adding to a model that already uses outdoor temperature. Put the steps in the order you would carry them out.

- (A) Fit the one-predictor model and note its R^2
- (B) Add the occupancy dummy and refit
- (C) Compare adjusted R^2 across the two models to see whether the new predictor earned its place
- (D) Check the residual plot of the model you intend to keep before trusting its predictions

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Q11. In `smf.ols("Cooling_kWh ~ Outdoor_Temp_F + BH", data=df).fit()`, what do the ~ and the + mean?

- A) ~ means 'is modelled as' with the response on the left; + adds a predictor
- B) ~ is approximate equality and + sums the two predictor columns arithmetically
- C) ~ negates the response and + concatenates the two column names into one
- D) Both are decorative; statsmodels reads the column order in the dataframe instead

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Q12. `model = smf.ols(...).fit()`. How do you get the fitted coefficients as numbers you can compute with?

- A) `model.summary()`
- B) `model.params`
- C) `model.coef()`
- D) `model.values`

Q13. In a statsmodels formula, what is the difference between `A*B` and `A:B`?

- A) `A*B` is the interaction alone, while `A:B` includes the main effects too
- B) `A*B` multiplies the two columns before fitting and `A:B` divides one by the other
- C) `A*B` expands to A + B + their interaction; `A:B` is the interaction alone
- D) They are identical, and statsmodels treats the two spellings the same way

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Q14. Adding a predictor always raises R^2, so models are compared by AIC as well. How do you read AIC?

- A) Higher is better, because AIC rewards how well the model fits the data
- B) Lower is better; it rewards fit but charges for each extra parameter
- C) AIC must fall between 0 and 1, the same way R^2 does, to be readable
- D) AIC is only comparable across different datasets, never within one

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Q15. `model.conf_int()` returns two columns for each coefficient. What are they, and what does it mean if the BusinessHours row spans zero?

- A) The lower and upper 95% confidence limits; spanning zero means no detectable effect
- B) The minimum and maximum fitted values; spanning zero means the model predicts negatives
- C) The standard error and the t-statistic; spanning zero means the data is collinear
- D) The first and last coefficient estimates across the iterations of the fit